

Study on Time and Light as Non-Newtonian Fluids

Introduction

This study aims to develop a mathematical equation that unequivocally establishes the possibility of creating a model to define the ideal conditions for considering time as a fluid. Once these conditions are met, the next goal is to corroborate, through appropriate equations, the behavior of time as a non-Newtonian fluid.

This will not be an easy, quick, or inexpensive task. It is important to acknowledge at the outset that there is no existing mathematical model within Einstein's general theory of relativity that explicitly treats time as a fluid in a literal sense. However, there are various analogies and extensions that relate time or spacetime to fluid properties. These ideas typically arise in theoretical contexts aiming to extend general relativity or connect it to quantum theories.

Below are some relevant mathematical approaches:

1. Fluid Analogies of Spacetime in General Relativity

In general relativity, spacetime can be considered a "dynamic medium" that deforms under the influence of energy and matter, described by Einstein's field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G c^{-4} T_{\mu\nu}$$

where $T_{\mu\nu}$ (the energy-momentum tensor) represents the energetic content of spacetime, analogous to how a fluid has density and pressure.

Although time itself is not treated as a fluid, the behavior of spacetime as a whole under these equations exhibits dynamics that are sometimes modeled with fluid analogies.

2. Analog Gravity and Fluid Models

In certain theoretical experiments, fluid models have been used to simulate spacetime behavior. For example:

- **Superfluid Models:** Spacetime is treated as a superfluid in some models, which could explain properties such as cosmic expansion or the formation of singularities.
- **Acoustic Analogies:** Acoustic black holes, where a fluid flows at supersonic speeds, have been used as analogs to study event horizons.

3. Hydrodynamic Models in Cosmology

In relativistic cosmology, some aspects of the universe are modeled using equations from relativistic hydrodynamics. For example:

- The **Friedmann equations**, which describe the universe's expansion, are derived under the assumption that the universe's content behaves as a homogeneous and isotropic fluid.
- In this context, time is viewed as an emergent dimension influenced by spacetime dynamics.

4. Emergent Theories of Spacetime

Theories attempting to merge general relativity with quantum mechanics, such as **loop quantum gravity**, describe spacetime as discrete and composed of "atoms of spacetime." In certain models, these microscopic units could behave similarly to fluid particles.

5. Fluid-Inspired Mathematical Equations

Some equations use structures analogous to the **Navier-Stokes equations** (from fluid mechanics) to describe spacetime dynamics. For instance:

$$\partial_\mu T^{\mu\nu} = 0$$

Here, $T^{\mu\nu}$ is interpreted as the energy-momentum tensor, and its properties can be compared to those of a relativistic fluid.

Based on these ideas, it is conceivable to develop or approximate a mathematical model that describes the conditions under which time behaves as a fluid and, furthermore, as a non-Newtonian fluid.

While achieving this may seem far-fetched, the possibility of finding the threshold where time ceases to be linear and curves to alter spacetime reality provides the necessary incentive to pursue this investigation.

Some studies explore the analogy between spacetime and a fluid, analyzing the conditions under which spacetime could exhibit such behavior. This perspective suggests that, at microscopic scales, spacetime might have a discrete structure, and its emergent continuous and geometric behavior could resemble that of a fluid in hydrodynamics.

For example, an article published by Europa Press discusses the possibility of treating spacetime as a fluid, where general relativity would act as an analogy to fluid

hydrodynamics. This approach implies that at very high energies, violations of Einstein's special relativity might occur, suggesting a more fundamental underlying structure of spacetime.

Additionally, recent research has developed analog models to describe gravitational phenomena using concepts from fluid dynamics. For instance, a study proposes a model to describe stationary black holes through the flow field of a spacetime-fluid around a three-dimensional point sink.

These approaches aim to better understand the nature of spacetime and the conditions under which it could exhibit fluid-like properties, providing a novel perspective in theoretical physics and cosmology.

Historical Background

Several prior studies have explored paradigms with significant similarities to those proposed in this study. Below are some recent approaches:

1. Spacetime Analogies with Fluids

- **Spacetime as a Superfluid:** Some physicists have suggested that at extremely small scales, spacetime could behave similarly to a superfluid, a phase of matter that flows without viscosity. This analogy seeks to reconcile general relativity with quantum mechanics, proposing that spacetime might have a discrete structure at the Planck scale (approximately 10^{-33} cm).
- **Liquid Spacetime Models:** Theoretical investigations have explored the possibility that spacetime resembles a fluid, implying that at microscopic scales, it is not continuous but discrete. This perspective could offer new pathways for developing a quantum theory of gravity.

2. Quantum Effects on the Perception of Time

- **Time Flow in the Quantum World:** Recent studies indicate that in the quantum realm, time does not necessarily progress in a single direction. Quantum systems can evolve simultaneously forward and backward in time, challenging our classical understanding of the temporal arrow.

3. Philosophical and Scientific Interpretations of Time

- **Time as an Illusion:** Some approaches in modern physics suggest that the flow of time could be an illusion derived from our perception and that past,

present, and future coexist simultaneously. This interpretation challenges the traditional notion of time progressing linearly.

4. Hydrodynamics Applications in Cosmology

- **Fluidity in Curved Spacetime:** Research has demonstrated the compatibility of certain fluid dynamics theories with general relativity, even in scenarios where spacetime is curved, such as in the fusion of neutron stars.

These perspectives offer innovative approaches to understanding the nature of time and space, though many of these ideas remain theoretical and require further experimental validation.

Theoretical Framework

To design a model that demonstrates the proposed hypothesis, it is necessary to correlate some principles of quantum physics and thermodynamics.

However, it is essential to clarify from the outset that a significant paradigm exists:

“Time is not literally considered a fluid in contemporary physics; some theories and models use fluid analogies to describe properties of spacetime and its relationship with quantum gravity.”

Therefore, to achieve a mathematical model validating the hypothesis of time as a fluid, at least four relevant approaches related to this idea must be considered:

1. Fluid Analogies in the Philosophy of Time

- **"The Flow of Time":** In philosophy, time is often described as a "flow" to capture our subjective experience of events moving from past to future. However, this is a metaphor and does not imply that time is literally a fluid.

2. Simulations in Quantum Gravity Theories

- Some theoretical models in quantum physics and general relativity consider spacetime as a "medium" that can behave analogously to a fluid.
- For example, in loop quantum gravity (LQG), spacetime could consist of discrete "atoms," and at certain scales, its properties might resemble those of a continuous fluid.

3. Hydrodynamic Equations Applied to Spacetime

- Studies of fluid analogs to general relativity have examined physical systems where fluid waves behave similarly to gravitational waves in spacetime. For example:
 - Research on black hole analogs using supersonic fluid flows, such as water black holes.

4. Mathematical Metaphors in Fluid Mechanics

- Some mathematical theories explore models where time equations (in cosmological or thermodynamic contexts) have formal similarities to fluid equations. For example, in the study of thermal time hypothesis, time is proposed to emerge as a parameter associated with thermal states.

To gain scientific validation for this study, each of these proposals must be pursued until a sufficiently sustainable model is developed to advance to the next phase.